

Submission to the Office of Science and Technology Policy (OSTP) and NITRD NCO

Request for Information: Development of an Artificial Intelligence (AI) Action Plan

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Introduction

As an independent researcher with expertise in AI systems and cybersecurity, I am submitting this response to the RFI to propose a priority policy action for the AI Action Plan: the development of a **Resilient AI Security Model (RASM)** for consumer desktops and laptops. This model leverages well-defined x86-based hardware running the Windows operating system, integrates a specialized system-on-chip (SoC) for real-time security, and pairs with a proactive monitoring AI to ensure self-healing and resilience. This proposal aligns with the Executive Order's goals of sustaining U.S. AI dominance and fostering private sector innovation while addressing national cybersecurity objectives.

Priority Action: Develop and Deploy a Resilient AI Security Model (RASM)

I recommend the inclusion of the following policy action in the AI Action Plan:

Fund and incentivize the creation of a Resilient AI Security Model (RASM) designed for x86-based consumer desktops and laptops running Windows, featuring a hardware-limited AI with real-time attack surface monitoring, a dedicated SoC for security, and a companion monitoring AI for proactive resilience.

Key Components of the Proposal

1. Hardware-Specific AI Design

- The RASM will be engineered for x86-based systems with known I/O and PCI internal components (e.g., standardized motherboards, GPUs, and storage devices). By limiting the AI's scope to a well-defined hardware ecosystem, it can achieve deep integration and optimization, reducing complexity and enhancing reliability.
- Rationale: A hardware-specific approach leverages the widespread use of x86 architecture in consumer devices (desktops and laptops), ensuring broad applicability while enabling precise control over the system's attack surface.

2. Comprehensive Attack Surface Awareness

- The AI will be trained to understand all attack surface points within the Windows OS and underlying hardware, including kernel-level processes, driver interactions, and firmware vulnerabilities.

- It will monitor and respond to anomalies in real time at both the hardware and OS layers (e.g., unauthorized memory access, malicious driver injections, or hardware-level exploits).
 - Rationale: Windows dominates the consumer OS market, yet its complexity creates numerous vulnerabilities. An AI with full visibility into these layers can preemptively thwart attacks, strengthening national cybersecurity.
3. **System-on-Chip (SoC) for Security**
- A dedicated SoC will be integrated into x86 systems to run the RASM independently of the main CPU. This SoC will handle encryption, secure boot processes, and real-time threat detection, isolating security functions from potential OS compromises.
 - Rationale: By offloading security to a separate chip, the system remains protected even if the OS or primary hardware is breached, mirroring successful strategies in mobile devices (e.g., Apple's T2 chip).
4. **Monitoring AI for Proactive Resilience**
- A secondary monitoring AI, hosted externally (e.g., in the cloud or on a separate device), will communicate with the SoC to analyze patterns, predict threats, and trigger self-healing actions (e.g., rolling back malicious changes, isolating compromised components).
 - This companion AI will use aggregated data from deployed RASM systems to improve resilience across the ecosystem.
 - Rationale: Proactive self-healing reduces downtime and damage, while collective learning enhances the model's effectiveness, supporting both individual users and national security.
5. **Consumer and National Cybersecurity Benefits**
- Targeted at consumer desktops and laptops, the RASM will protect millions of U.S. households and small businesses, which are frequent targets of cyberattacks.
 - It aligns with national cybersecurity objectives by reducing the attack surface of critical infrastructure endpoints, preventing consumer devices from becoming vectors in larger-scale attacks (e.g., botnets).
 - Rationale: Securing consumer endpoints strengthens the overall cybersecurity posture of the nation, a key component of U.S. AI leadership.

Policy Recommendations

To implement this action, I propose the following steps:

1. **Federal Funding and R&D Grants:** Allocate resources to develop the RASM, including partnerships with x86 hardware manufacturers (e.g., Intel, AMD) and Microsoft for Windows integration.
2. **Private Sector Incentives:** Offer tax credits or subsidies to companies producing RASM-enabled devices, encouraging rapid adoption in the consumer market.
3. **Regulatory Sandbox:** Establish a testing environment where developers can deploy and refine the RASM without immediate regulatory burdens, fostering innovation.

4. **Standards Development:** Collaborate with industry to define open standards for the SoC and AI communication protocols, ensuring interoperability and scalability.
5. **Public Awareness Campaign:** Educate consumers on the benefits of RASM-enabled devices to drive demand and adoption.

Why This Matters

This proposal advances U.S. AI dominance by pioneering a hardware-AI synergy that competitors like China have yet to match in the consumer space. It avoids burdensome regulation by embedding security into the design process, leaving manufacturers free to innovate within a secure framework. Moreover, it directly supports national cybersecurity by hardening consumer devices against exploitation, a critical need given the rise in state-sponsored and criminal cyberattacks.

Conclusion

The Resilient AI Security Model (RASM) offers a forward-thinking, practical solution to bolster U.S. AI leadership and protect the nation's digital ecosystem. By integrating hardware-specific AI, a secure SoC, and proactive resilience into consumer devices, this action ensures innovation thrives without compromising security. I urge OSTP and NITRD NCO to prioritize this initiative in the AI Action Plan, leveraging America's strengths in x86 hardware and software to set a global standard for resilient AI systems.

Thank you for considering this submission. I am available to provide further details or collaborate on implementation strategies.

Sincerely,

Zachary Lewis